

Master Thesis Review

Conversational Explanations Using Large Language Models
on Top of the Explainable AI Question Bank

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Dear Team of the Examination Office at the Department of Mathematics and Computer Science,

Henry Khalil El-Jawhari's Master's Thesis "Conversational Explanations Using Large Language Models on Top of the Explainable AI Question Bank" explores the integration of large language models (LLMs) as conversational explanatory agents within the field of explainable AI (XAI). The work focuses on bridging the gap between XAI interfaces and the rigorous evaluation of LLM-generated responses, specifically within the high-stakes domain of healthcare.

The thesis addresses the critical need for transparency and trust in AI systems by using LLMs to transform complex technical details into understandable narratives. El-Jawhari introduces a new Accuracy-Completeness-Faithfulness (ACF) composite metric and a readability benchmark to evaluate conversational agents. The framework employs an expanded XAI Question Bank (XAI-QB) to assess the ability of LLMs to convey the architecture and performance of an AI system to domain experts.

The thesis is logically structured and written with great scientific clarity. It progresses from a comprehensive theoretical foundation to a rigorous empirical evaluation. After conducting a thorough literature review of human-centered AI (HC-AI) and explainable AI (XAI), El-Jawhari outlines the methodology behind his vaccine uptake prediction tool and subsequent benchmarking of LLMs. His formal, precise writing style effectively integrates technical, mathematical formulations, such as accumulated local effects and Shapley values, with practical software artifacts, such as the JSON-based instructional template for clinical prompting. This balance of theoretical rigor and applied technical documentation, supported by extensive references to current AI research, ensures the work is reproducible and accessible to a multidisciplinary audience.

The robust methodology involves creating an AI prediction tool to simulate a real-world healthcare scenario. A selection of state-of-the-art models, including ChatGPT, Claude, and Gemini, were evaluated against human reference answers. The research rigorously benchmarks these models across the defined ACF dimensions and finds that, while LLMs perform comparably well, specific models exhibit distinct strengths. For instance, ChatGPT is consistent in its accuracy, while Gemini is superior in its readability. This work makes three key contributions: 1) the ACF Composite Metric, a unified, reproducible quality score that combines accuracy, completeness, and faithfulness; 2) Readability Benchmarking, which uses human reference answers to determine the ideal



balance between technical detail and ease of understanding; and 3) an instructional template with a reusable framework for fine-tuning LLM prompts to act as clinical explanatory agents.

While the thesis validates LLMs as effective XAI tools, it also identifies areas for future refinement. Specifically, the ACF score reveals that some XAI-QB questions require greater specificity to prevent "hallucinations" or vague responses (512, 599). Furthermore, the work emphasizes the necessity of "human-in-the-loop" verification to prevent clinical misuse caused by placing too much trust in readable yet inaccurate explanations.

Overall, this thesis is a praiseworthy and original contribution to the intersection of human-centered AI and natural language processing. It offers actionable insights for designing more transparent and trustworthy AI systems. **For the written thesis, I would confidently give Mr. H. K. El-Jawhari a grade of 1.0 for his very good and independently accomplished work.**

Potential defense questions:

1. How do you envision the trade-off between model "creativity" in conversational agents and the strict "faithfulness" required in clinical settings?
2. Your results show that Gemini leads in readability, while ChatGPT leads in accuracy. How would you design a hybrid system to maximize both?
3. How might the ACF metric be adapted for real-time evaluation in a live clinical interface?
4. What are the primary ethical risks when an LLM provides a "highly readable" yet factually incorrect explanation to a medical professional?
5. How could your instructional template be generalized to non-healthcare domains, such as legal or financial AI systems?

Best regards,

A handwritten signature in black ink, appearing to be 'G. Hattab'.

PD Dr Georges Hattab
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